

Column Name	Source Dataset(s)	Reason for Inclusion			
enrollment_id	Generated during ETL	(Primary Key) This is a new, unique ID created for every single row in this table. It ensures that each specific application by a learner to an opportunity can be uniquely identified, which is critical for accurate analysis and data management.			
learner_id	Learner_Raw, Cognito_Raw2	(Foreign Key) This is the unique identifier for the student. It's the central pillar that connects a person's demographic information (from Cognito) with their academic background (from Learner) and their actions (from LearnerOpportunity).			
opportunity_id	Opportunity_Raw, LearnerOpportunity_Raw	(Foreign Key) This identifies the specific opportunity the learner applied for. It allows us to link to the opportunity's details, such as its name and category, and analyze which opportunities are most popular.			
cohort_id	CohortRaw, LearnerOpportunity_Raw	(Foreign Key) This identifies the specific cohort associated with the learner's application. It connects the learner to a specific timeline, including the cohort's start and end dates.			
email	Cognito_Raw2	Provides essential contact information and serves as a common, human-readable identifier for the learner.			
gender	Cognito_Raw2	Basic demographic information used for high-level segmentation and analysis (e.g., analyzing application trends across different genders).			
birthdate	Cognito_Raw2	Allows for age-based analysis, such as understanding the age distribution of applicants for certain programs.			
city, state, country	Cognito_Raw2, Learner_Raw	Key geographic data used to analyze regional trends, identify target markets, and understand the geographic distribution of learners.			
degree, institution, major	Learner_Raw	Provides the learner's academic background. This is crucial for analyzing which educational backgrounds are most correlated with success in certain opportunities.			
opportunity_name, opportunity_category	Opportunity_Raw	Gives descriptive context to the opportunity_id. Instead of just seeing an ID, analysts can immediately see the name and type of opportunity, making reports much easier to read and understand.			
cohort_start_date, cohort_end_date	CohortRaw	Provides the specific timeline for the learner's program. This is essential for resource planning, progress tracking, and analyzing program duration.			
apply_date	LearnerOpportunity_Raw	A critical piece of data that provides the exact timestamp of the application. It allows for time-series analysis to track application velocity and trends.			
status	LearnerOpportunity_Raw	Shows the outcome of the application (e.g., 'Accepted', 'Rejected', 'In Progress'). This is often the primary metric used to measure the success of both the learner and the program.			

Keys and Relationships: The Table's Skeleton

- Primary Key (PK):** `enrollment_id`
 - It's a unique ID for every single row in the master table.
 - It's essential because a learner can apply multiple times; this key distinguishes each unique application.
 - It guarantees data integrity and prevents confusion.
- Foreign Keys (FKs):** `learner_id`, `opportunity_id`, `cohort_id`
 - These keys act as the "glue" that links data back to the original source tables.
 - They enforce a critical rule: you can't have an application for a learner or opportunity that doesn't exist.
 - This prevents "orphan records" and ensures all data is valid and connected.

Overall Reasoning for This Design

- Creates a Single Source of Truth:** Combines five fragmented files into one complete narrative for each student.
- Enables Complex Analysis:** Allows you to easily answer detailed questions that cross multiple datasets (e.g., "What is the academic background of applicants for X opportunity?").
- Provides a Robust & Scalable Structure:** The design is strong, can grow with more data, and is perfectly optimized for building dashboards and reports.